Autonomous Robotic Fruit Harvesting System

Vision-Guided Mobile Platform with Hybrid Robotic Workspace

Existing Solutions

Manual fruit harvesting

Labour-intensive, variable performance, and fatigue-prone.

Mechanical harvesting machines

Broad coverage but limited selectivity and high impact on fruit and plants.

Fixed robotic platforms

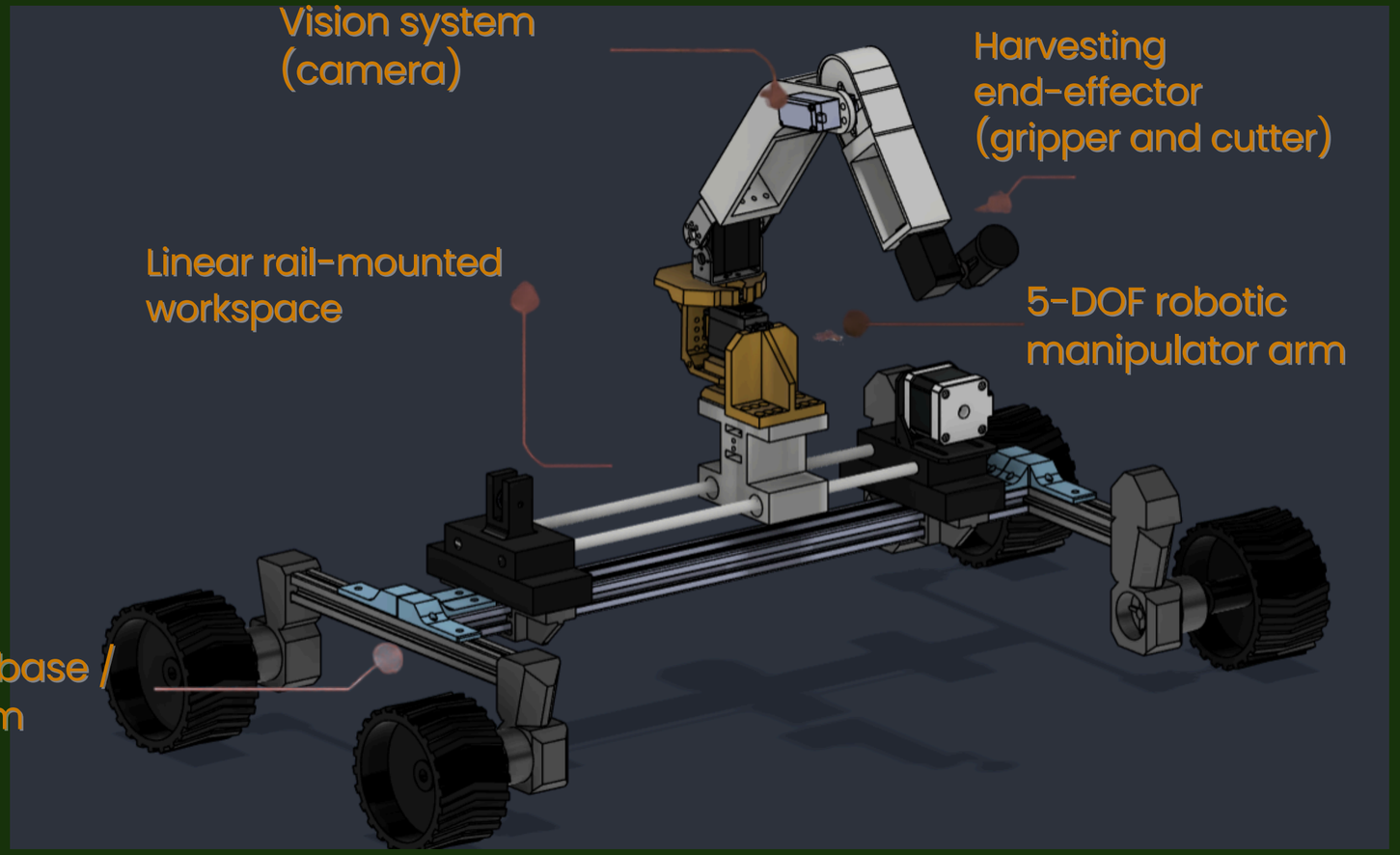
Stable operation but limited adaptability to varying tree structures and row layouts.

Mobile robots with restricted reach

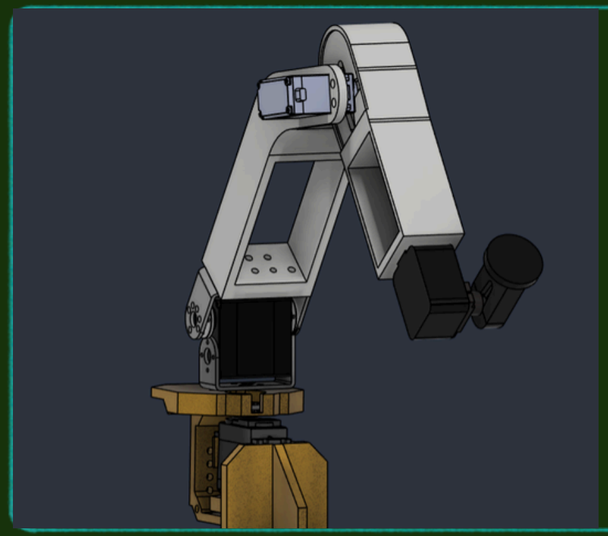
Mobility enables access, but limited manipulator reach reduces effective coverage.

Proposed Solution

An integrated autonomous system that combines mobile navigation, precision manipulation, and vision intelligence to perform safe, reliable fruit harvesting in complex orchard environments.

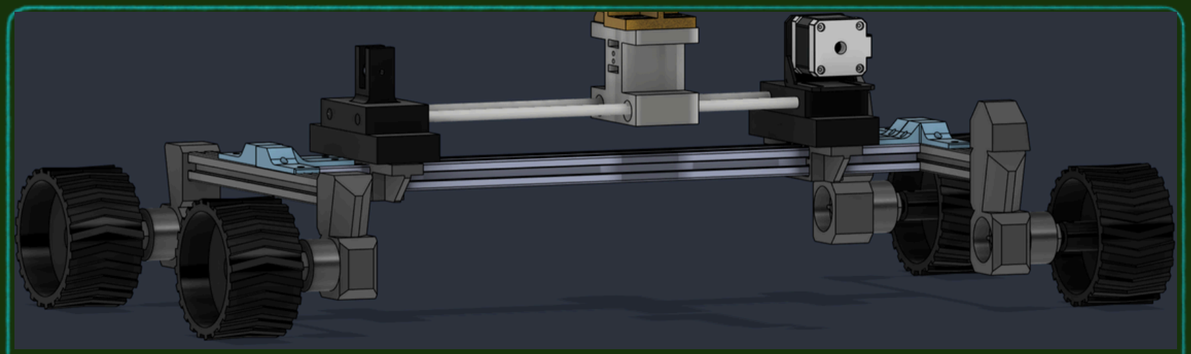


Innovation and Novelty



Hybrid Robotic Workspace

Linear rail combined with 5-DOF arm for extended reach



Vision-Guided Mobile Platform

Camera-based fruit detection and autonomous navigation



Custom-Trained YOLO Model

Fruit detection and ripeness classification with dedicated dashboard



Modular wrist

Precision gripping, detachment, and fruit collection

Hybrid Workspace





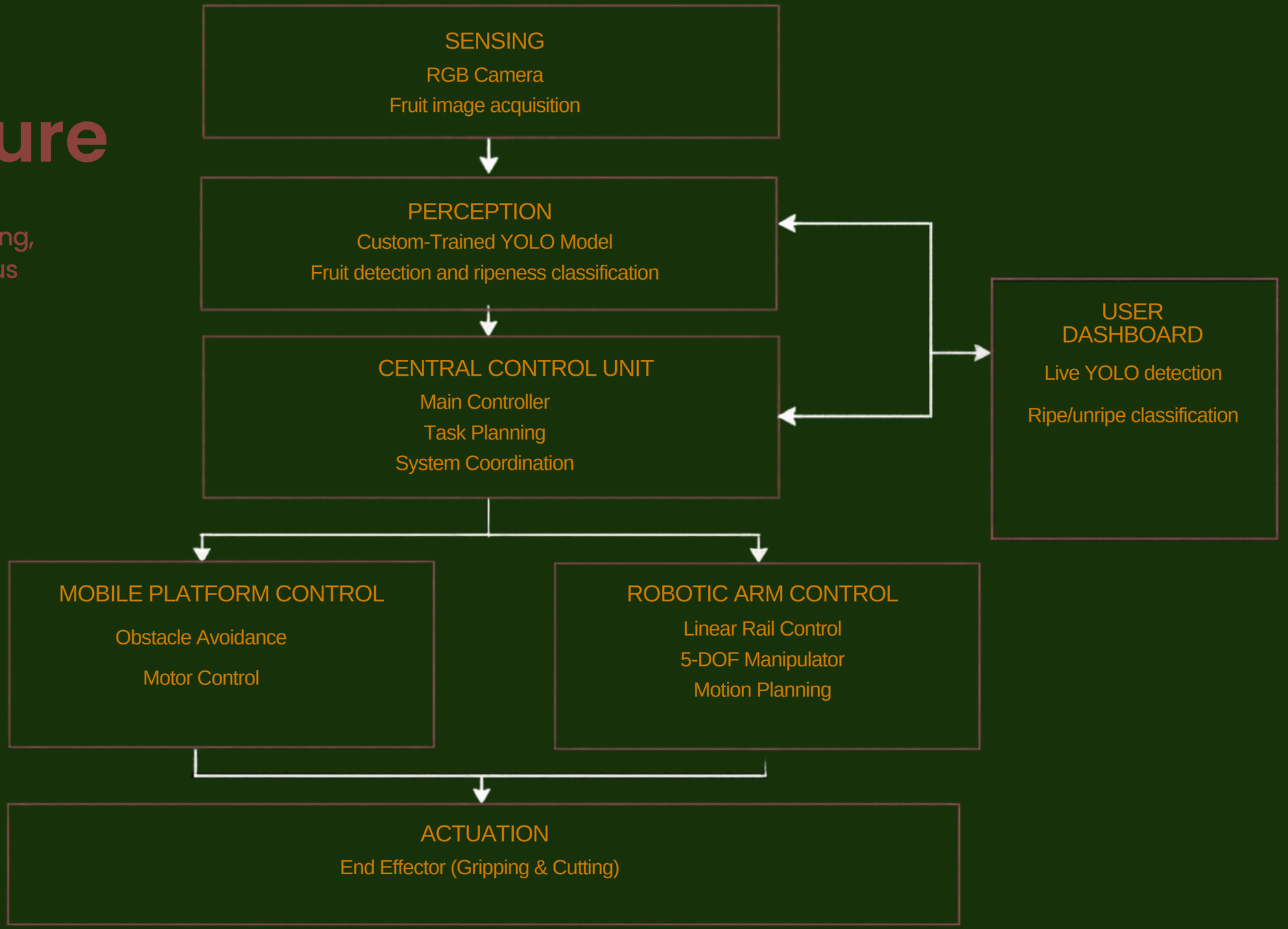






System Architecture

Integrated perception, planning, and actuation for autonomous fruit harvesting.



Objectives

01 Develop an autonomous mobile platform for agricultural navigation

02 Design a hybrid robotic arm integrating linear rail and 5-DOF manipulator

03 Train a custom YOLO model for fruit detection and ripeness classification

04 Develop a real-time fruit detection dashboard

05 Implement vision-guided fruit localization and positioning



Industrial Applications

- Commercial Fruit Orchards and Plantations
- Greenhouse and Polyhouse Farming
- Precision and Smart Agriculture
- Large-Scale Agricultural Farms
- Horticulture and Crop Management
- Food Processing and Sorting Industries
- Export-Oriented Fruit Production
- Agricultural Research and Development
- Multi-Crop Harvesting Automation
- Autonomous Farming Equipment

